

for mathematical functions and has a state-of-art compiler, which generates the optimised code with very high levels of numerical precision. Julia has the inherent capability to execute code in the parallel mode.

The development of Julia has been spearheaded by Jeff Bezanson, Stefan Karpinski, Viral Shaw and Alan Edelman. The initial version was released in 2012. The Julia community is on the rise, which is very evident from the official GitHub page of Julia (<https://github.com/JuliaLang/julia>). The current version of Julia is 0.4.6, as of June 19, 2016.

Julia installation

Julia can be installed in all major operating systems such as Linux, Mac OS X and MS Windows. The latest version of Julia may be downloaded from <http://julialang.org/downloads/>, where 32- and 64-bit versions are provided for all platforms. You need to select the version matching your operating system and configuration. Installation is straightforward. If you are using Windows, download the Julia set-up file, extract and execute. If you are a Mac OS X user, the corresponding 'dmg' file may be downloaded and executed. For Linux users, the distribution specific instructions are provided in <http://julialang.org/downloads/platform.html>.

In this article, the Ubuntu installation is provided as a sample. Execute the following commands for Julia installation in Ubuntu:

```
sudo add-apt-repository ppa:staticfloat/juliareleases
sudo add-apt-repository ppa:staticfloat/julia-deps
sudo apt-get update
sudo apt-get install julia
```

On successful installation, type 'Julia' at the terminal to start Julia. The welcome screen of Julia is shown in Figure 1.

Julia execution modes

Julia can be executed in three different ways as illustrated in Figure 2.

- **Command line execution:** To work in this mode, just download Julia as mentioned above and start working at the terminal by typing `julia`.
- **Using the Integrated Development Environment Juno:** To work in this mode, first the Julia command line needs to be downloaded. If *Atom* is not installed in your system, then get it from <https://atom.io/>. After the successful installation of Atom, navigate to the *Settings::Install* panel (press `Ctrl` to open *Settings*). Enter `uber -juno` and install the same.
- **Using JuliaBox.com.** No installation is required for this mode. You can visit this URL and start interacting with Julia.

The features of Julia

The significant features of Julia are illustrated in Figure 3. This language has the ability to define function behaviour

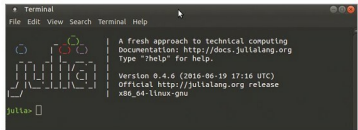


Figure 1: Julia welcome screen



Figure 2: Julia execution modes



Figure 3: The features of Julia

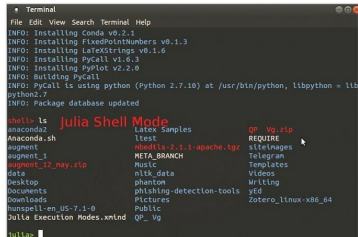


Figure 4: Julia shell mode

across many combinations of parameter types. This feature is known as *Multiple Dispatch*. Availability of an in-built package manager, attractive REPL, the ability to execute code in a parallel and distributed environment, and optimal code generation are the other noteworthy features of Julia.

Julia REPL

Julia provides an attractive REPL (*Read – Evaluate – Print – Loop*), which facilitates the execution of Julia expressions. The expressions are evaluated immediately and the results are printed.

```
julia> 5 + 3
8
```